Technology Stack (Architecture & Stack)

Date	08 October 2026
Team ID	_____
Project Name	Import Data using Transform Maps (Spreadsheet)
Maximum Marks	4 Marks

Technical Architecture:

The deliverable includes the architecture of the solution and the information as per Table-1 and Table-2. The architecture flow is: Excel spreadsheet -> Data Source -> Import Set Table -> Transform Map (field maps, coalesce) -> Target Table -> Reports -> Dashboard.

Table-1 : Components & Technologies

S.No	Component	Description	Technology
1.	User Interface	How the user interacts with the application: ServiceNow web interface for loading data, running transforms and viewing dashboards	ServiceNow Next Experience (web UI)
2.	Application Logic-1	Field mapping from spreadsheet columns to target table fields	ServiceNow Transform Map, Field Maps
3.	Application Logic-2	Avoid duplicate records using the coalesce option on a unique field	Transform Map - Coalesce
4.	Application Logic-3	Validation and transformation of data (optional transform scripts)	JavaScript (Transform Scripts)
5.	Database	Staging data in the Import Set table; final data in the custom target table	ServiceNow relational database (tables and fields)
6.	Cloud Database	Database service on cloud	ServiceNow cloud instance
7.	File Storage	Excel file stored as an attachment on the Data Source record	ServiceNow attachments
8.	External API-1	Not used in this project	N/A
9.	External API-2	Not used in this project	N/A
10.	Machine Learning Model	Not used in this project	N/A
11.	Infrastructure (Server / Cloud)	Application deployed and run on the ServiceNow cloud instance Local: Excel prepared on the user's computer Cloud: ServiceNow instance	ServiceNow Cloud

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	No open-source framework is needed; the solution uses built-in platform features	ServiceNow platform (no-code / low-code)
2.	Security Implementations	Role-based access for import and administration; ACLs on the target table	ServiceNow roles, ACLs
3.	Scalable Architecture	Staging table and reusable transform map handle larger files without redesign	Import Set + Transform Map
4.	Availability	Hosted on the ServiceNow cloud, accessible from any browser	ServiceNow Cloud instance

S.No	Characteristics	Description	Technology
5.	Performance	Bulk rows are transformed in one run; staging table keeps source data separate from live data	Import Sets, Transform Maps

References:

https://docs.servicenow.com (Import Sets and Transform Maps)
https://c4model.com/